DIVINECODES / PROJECT 2

Where is that car's odometer right now?

Delayed readings make valuation and defleet decisions uncertain.

27,630

development vehicles

2,699,762 supplied status rows

One operational workflow

Estimate today.
Forecast the next decision.
Review suspicious reports.

THE INSIGHT

A timestamp is not always a fresh odometer reading.

Two clocks, one reading

Measurement date: when the mileage belongs.
Availability date: when a prediction may first use it.

Retiming stale handover values helps, but future-recorded rental rows must stay hidden.

424,008 paired vehicle-days have lower OnRent mileage.
14,660 have higher mileage: the pattern needs qualification.

Aggregate fleet evidence; no individual vehicle records shown.

THE METHOD

Predict with the information available at the time.

1 · Establish the timeline

Retimed readings + recorded availability for mileage, rentals and category.

2 · Compare simple and rental-aware models

Choose on validation; calibrate ranges on a separate later period.

3 · Deliver an estimate and a review queue

Per-vehicle output, age of last reading, uncertainty and explainable flags.

Stable vehicle holdout + later periods. No LLM or paid inference. Future-field mutation tests protect the boundary.

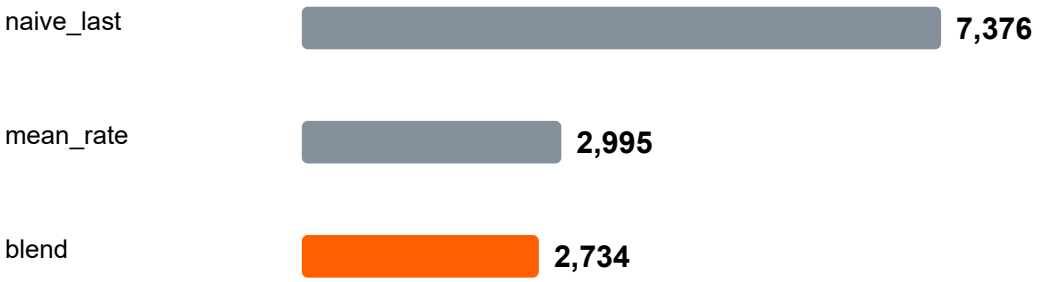
HELD-OUT RESULTS · MAE KM · LOWER IS BETTER

Measured against two baselines.

Nowcast · 20,732 targets



Forecast · 14,274 targets



26.0% lower nowcast MAE vs last reading · **62.9%** lower forecast MAE

Validation-selected models shown in orange. Error-injection episode recall: 94.2%; exact-reading recall: 93.9%. Real flag precision is not yet known. Recorded target mileage may itself contain errors.

HOW SIXT COULD USE IT

A daily decision aid, with explicit uncertainty.

Valuation & defleet

Current and future mileage.
Reading age and prediction range.
Separate mileage-limit alerts.

Review operations

Largest discrepancies first.
Context before accusation.
Human confirmation before action.

Next: confirm status semantics, label real flags, and validate long-term rentals and range coverage in shadow mode.

Runnable local demo uses made-up vehicles. No proven fraud, monetary savings, production readiness or hidden-test accuracy is claimed.